1.

```
e2301760@e2301760-VirtualBox:~$ ip route

default via 192.168.73.254 dev enp0s3 proto dhcp src 192.168.73.155 metric 100
192.168.73.0/24 dev enp0s3 proto kernel scope link src 192.168.73.155 metric 100
```

2.

```
e2301760@e2301760-VirtualBox:~$ ping -c 1 -s 61 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 61(89) bytes of data.
69 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=12.7 ms

--- 8.8.8.8 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 12.708/12.708/12.708/0.000 ms
```

3.

```
e2301760@e2301760-VirtualBox:~$ netstat -an | grep LISTEN

tcp        0      0 0.0.0.0:11225           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:5072          0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:44703         0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN
tcp6       0      0 ::1:631                 :::*                    LISTEN
unix  2      [ ACC ]     STREAM     LISTENING     1852     /run/systemd/io.systemd.sy
sext
unix  2      [ ACC ]     STREAM     LISTENING     20124    /tmp/dbus-5Lz7FTTjRS
unix  2      [ ACC ]     STREAM     LISTENING     11914    /run/user/1000/systemd/pri
vate
unix  2      [ ACC ]     STREAM     LISTENING     13584    /run/user/1000/bus
unix  2      [ ACC ]     STREAM     LISTENING     7590     /run/systemd/resolve/io.sy
stemd.Resolve
unix  2      [ ACC ]     STREAM     LISTENING     7591     /run/systemd/resolve/io.sy
stemd.Resolve.Monitor
unix  2      [ ACC ]     STREAM     LISTENING     13585    /run/user/1000/gnupg/S.dir
mngr
unix  2      [ ACC ]     STREAM     LISTENING     13587    /run/user/1000/gcr/ssh
unix  2      [ ACC ]     STREAM     LISTENING     13589    /run/user/1000/keyring/con
trol
unix  2      [ ACC ]     STREAM     LISTENING     13591    /run/user/1000/gnupg/S.gpg
-agent.browser
unix  2      [ ACC ]     STREAM     LISTENING     13593    /run/user/1000/gnupg/S.gpg
-agent.extra
unix  2      [ ACC ]     STREAM     LISTENING     13595    /run/user/1000/gnupg/S.gpg
-agent
unix  2      [ ACC ]     STREAM     LISTENING     13597    /run/user/1000/gnupg/S.key
boxd
unix  2      [ ACC ]     STREAM     LISTENING     13599    /run/user/1000/pulse/nativ
e
unix  2      [ ACC ]     STREAM     LISTENING     13601    /run/user/1000/pipewire-0
unix  2      [ ACC ]     STREAM     LISTENING     13603    /run/user/1000/pipewire-0-
manager
unix  2      [ ACC ]     STREAM     LISTENING     13605    /run/user/1000/pk-debconf-
socket
unix  2      [ ACC ]     STREAM     LISTENING     13607    /run/user/1000/speech-disp
atcher/speechd.sock
unix  2      [ ACC ]     STREAM     LISTENING     13612    /run/user/1000/gnupg/S.gpg
-agent.ssh
unix  2      [ ACC ]     STREAM     LISTENING     8467     /run/avahi-daemon/socket
unix  2      [ ACC ]     STREAM     LISTENING     8469     /run/dbus/system_bus_socke
t
unix  2      [ ACC ]     STREAM     LISTENING     8470     /run/uuidd/request
unix  2      [ ACC ]     STREAM     LISTENING     13671    /run/user/1000/keyring/pkc
s11
unix  2      [ ACC ]     STREAM     LISTENING     12142    /run/user/1000/keyring/ssh
```

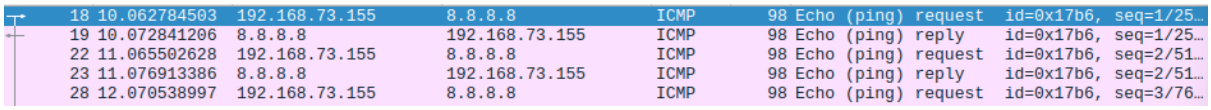
631 and 53 are well-known so there are 2

4.

```
e2301760@e2301760-VirtualBox:~$ traceroute 8.8.8.8

traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte pack
ets
 1  ifw.cc.puv.fi (192.168.73.254)  0.622 ms  0.906 ms  0.
590 ms
 2  efw.cc.puv.fi (193.166.141.17)  0.469 ms  0.921 ms  1.
275 ms
 3  uva-r1.cc.puv.fi (193.166.141.2)  0.935 ms  0.805 ms
0.992 ms
 4  vaasa1-xe-0-0-0-0-1.ip.funet.fi (193.167.253.72)  0.91
1 ms  0.671 ms  0.821 ms
 5  vaasa4.ip.funet.fi (86.50.255.107)  10.546 ms  10.183
ms  9.653 ms
 6  turku4-et-0-1-0-1.ip.funet.fi (86.50.254.236)  9.649 m
s  10.276 ms  9.798 ms
 7  stockholm4-et-0-0-0-1.ip.funet.fi (86.50.254.251)  9.5
23 ms  9.250 ms  10.715 ms
 8  se-sthb1.nordu.net (109.105.102.0)  10.432 ms  10.149
ms  9.721 ms
 9  se-bma2.nordu.net (109.105.101.93)  9.442 ms  9.903 ms
  9.423 ms
10  * * *
11  72.14.237.55 (72.14.237.55)  10.661 ms 216.239.57.151
(216.239.57.151)  10.216 ms 216.239.48.77 (216.239.48.77)
 11.576 ms
12  142.250.239.183 (142.250.239.183)  11.110 ms 172.253.7
2.123 (172.253.72.123)  10.798 ms 209.85.242.83 (209.85.24
2.83)  14.582 ms
13  dns.google (8.8.8.8)  10.151 ms  13.575 ms  10.699 ms
e2301760@e2301760-VirtualBox:~$ traceroute -n 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte pack
ets
 1  192.168.73.254  0.952 ms  0.968 ms  0.810 ms
 2  193.166.141.17  0.668 ms  0.561 ms  0.431 ms
 3  193.166.141.2  1.250 ms  1.114 ms  1.033 ms
 4  193.167.253.72  0.932 ms  0.855 ms  0.757 ms
 5  86.50.255.107  9.623 ms  9.439 ms  9.217 ms
 6  86.50.254.236  9.559 ms  9.430 ms  9.332 ms
 7  86.50.254.251  10.095 ms  9.993 ms  9.916 ms
 8  109.105.102.0  8.981 ms  9.391 ms  9.142 ms
 9  109.105.101.93  9.422 ms  9.334 ms  9.264 ms
10  * * *
11  216.239.57.151  16.400 ms 192.178.248.215  15.766 ms 2
16.239.48.77  15.512 ms
12  172.253.72.119  15.382 ms 209.85.242.11  16.792 ms 108
.170.233.41  16.647 ms
13  8.8.8.8  15.032 ms  14.787 ms  14.681 ms
e2301760@e2301760-VirtualBox:~$
```

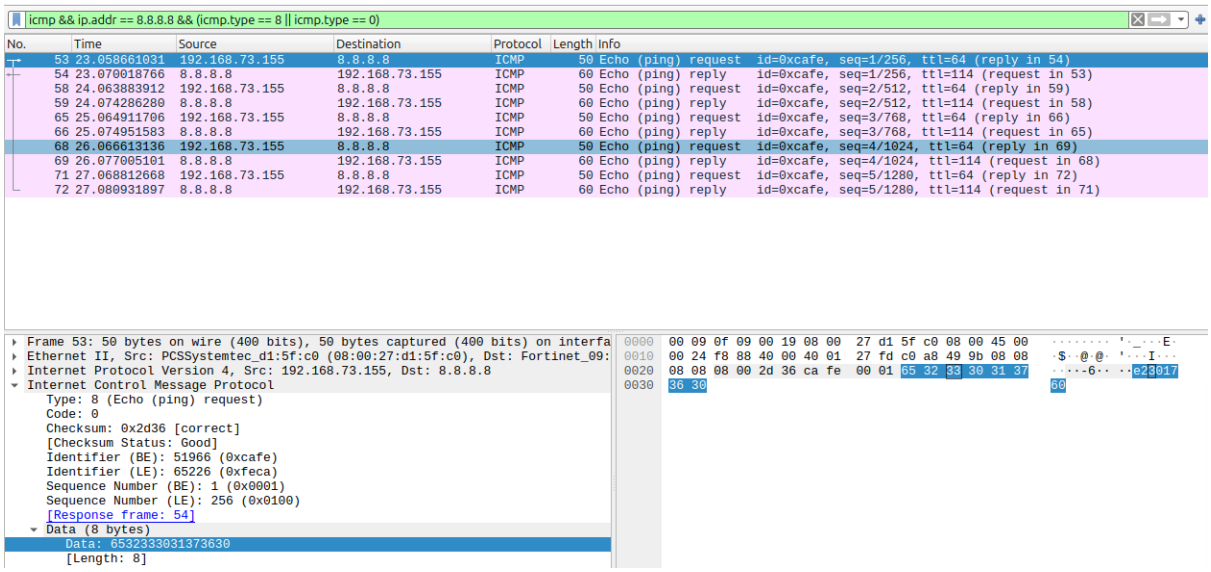
5.



18	10.062784503	192.168.73.155	8.8.8.8	ICMP	98	Echo (ping) request id=0x17b6, seq=1/25…
19	10.072841206	8.8.8.8	192.168.73.155	ICMP	98	Echo (ping) reply id=0x17b6, seq=1/25…
22	11.065502628	192.168.73.155	8.8.8.8	ICMP	98	Echo (ping) request id=0x17b6, seq=2/51…
23	11.076913386	8.8.8.8	192.168.73.155	ICMP	98	Echo (ping) reply id=0x17b6, seq=2/51…
28	12.070538997	192.168.73.155	8.8.8.8	ICMP	98	Echo (ping) request id=0x17b6, seq=3/76…

if I send 2 pings there are 4 packets captured

6.

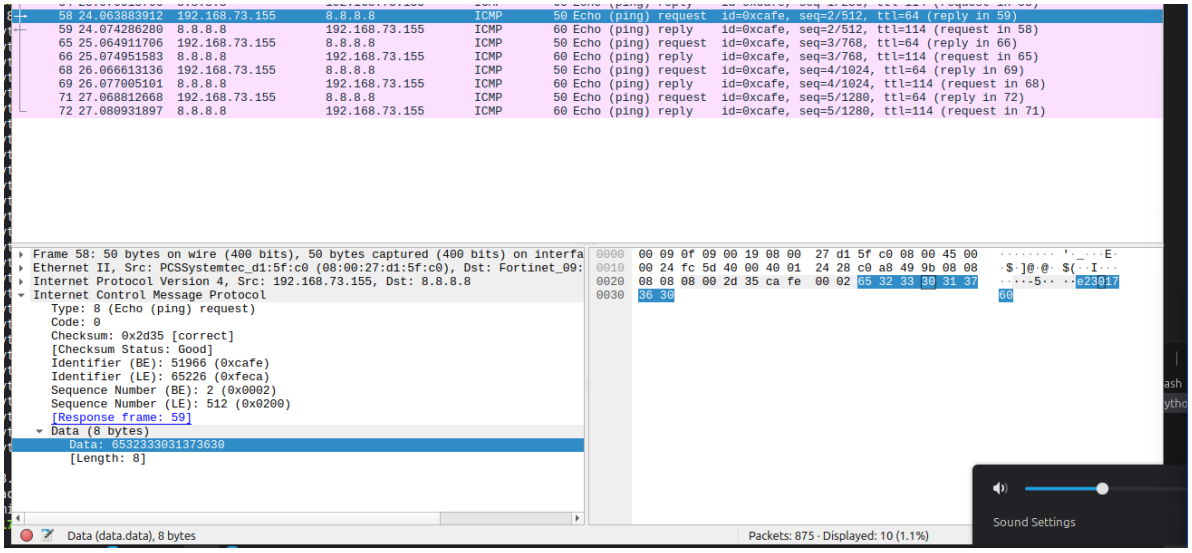


icmp && ip.addr == 8.8.8.8 && (icmp.type == 8 || icmp.type == 0)

No.	Time	Source	Destination	Protocol	Length	Info
53	23.058661031	192.168.73.155	8.8.8.8	ICMP	50	Echo (ping) request id=0xcafe, seq=1/256, ttl=64 (reply in 54)
54	23.070018766	8.8.8.8	192.168.73.155	ICMP	60	Echo (ping) reply id=0xcafe, seq=1/256, ttl=114 (request in 53)
58	24.063883912	192.168.73.155	8.8.8.8	ICMP	50	Echo (ping) request id=0xcafe, seq=2/512, ttl=64 (reply in 59)
59	24.074286280	8.8.8.8	192.168.73.155	ICMP	60	Echo (ping) reply id=0xcafe, seq=2/512, ttl=114 (request in 58)
65	25.064911706	192.168.73.155	8.8.8.8	ICMP	50	Echo (ping) request id=0xcafe, seq=3/768, ttl=64 (reply in 66)
66	25.074951583	8.8.8.8	192.168.73.155	ICMP	60	Echo (ping) reply id=0xcafe, seq=3/768, ttl=114 (request in 65)
68	26.066613136	192.168.73.155	8.8.8.8	ICMP	50	Echo (ping) request id=0xcafe, seq=4/1024, ttl=64 (reply in 69)
69	26.077005101	8.8.8.8	192.168.73.155	ICMP	60	Echo (ping) reply id=0xcafe, seq=4/1024, ttl=114 (request in 68)
71	27.068812668	192.168.73.155	8.8.8.8	ICMP	50	Echo (ping) request id=0xcafe, seq=5/1280, ttl=64 (reply in 72)
72	27.080931897	8.8.8.8	192.168.73.155	ICMP	60	Echo (ping) reply id=0xcafe, seq=5/1280, ttl=114 (request in 71)

```
▸ Frame 53: 50 bytes on wire (400 bits), 50 bytes captured (400 bits) on interfa
▸ Ethernet II, Src: PCSSystemtec_d1:5f:c0 (08:00:27:d1:5f:c0), Dst: Fortinet_09:
▸ Internet Protocol Version 4, Src: 192.168.73.155, Dst: 8.8.8.8
▾ Internet Control Message Protocol
    Type: 8 (Echo (ping) request)
    Code: 0
    Checksum: 0x2d36 [correct]
    [Checksum Status: Good]
    Identifier (BE): 51966 (0xcafe)
    Identifier (LE): 65226 (0xfeca)
    Sequence Number (BE): 1 (0x0001)
    Sequence Number (LE): 256 (0x0100)
    [Response frame: 54]
  ▾ Data (8 bytes)
      Data: 6532333031373630
      [Length: 8]

0000  00 09 0f 09 00 19 08 00  27 d1 5f c0 08 00 45 00   ····· '·_···E·
0010  00 24 f8 88 40 00 40 01  27 fd c0 a8 49 9b 08 08   ·$··@·@· '···I···
0020  08 08 08 00 2d 36 ca fe  00 01 65 32 33 30 31 37   ····-6·· ··e23017
0030  36 30                                              60
```



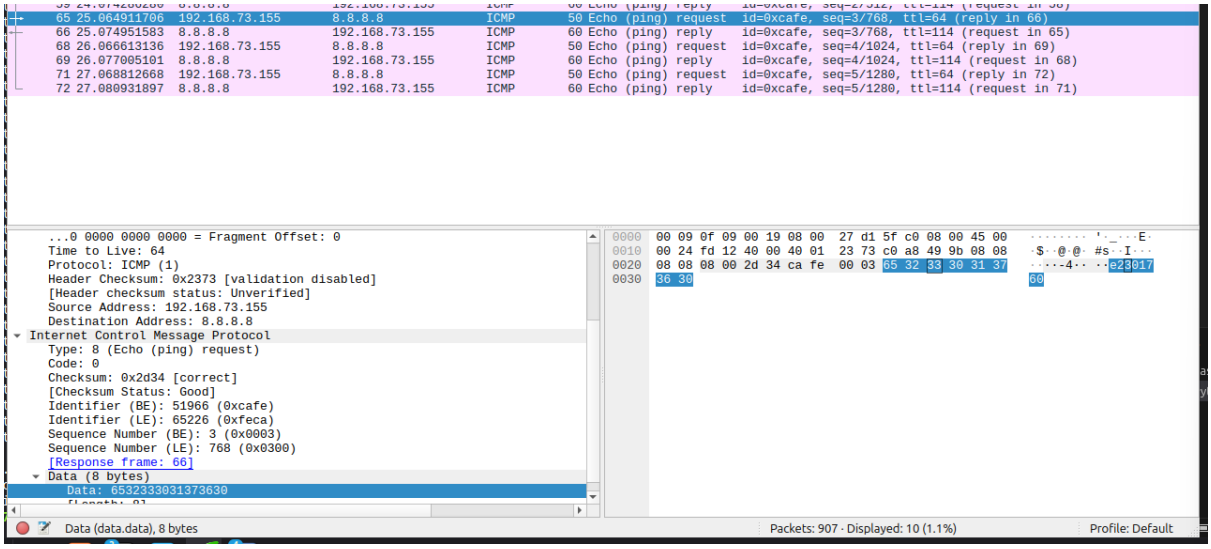
58	24.063883912	192.168.73.155	8.8.8.8	ICMP	50	Echo (ping) request id=0xcafe, seq=2/512, ttl=64 (reply in 59)
59	24.074286280	8.8.8.8	192.168.73.155	ICMP	60	Echo (ping) reply id=0xcafe, seq=2/512, ttl=114 (request in 58)
65	25.064911706	192.168.73.155	8.8.8.8	ICMP	50	Echo (ping) request id=0xcafe, seq=3/768, ttl=64 (reply in 66)
66	25.074951583	8.8.8.8	192.168.73.155	ICMP	60	Echo (ping) reply id=0xcafe, seq=3/768, ttl=114 (request in 65)
68	26.066613136	192.168.73.155	8.8.8.8	ICMP	50	Echo (ping) request id=0xcafe, seq=4/1024, ttl=64 (reply in 69)
69	26.077005101	8.8.8.8	192.168.73.155	ICMP	60	Echo (ping) reply id=0xcafe, seq=4/1024, ttl=114 (request in 68)
71	27.068812668	192.168.73.155	8.8.8.8	ICMP	50	Echo (ping) request id=0xcafe, seq=5/1280, ttl=64 (reply in 72)
72	27.080931897	8.8.8.8	192.168.73.155	ICMP	60	Echo (ping) reply id=0xcafe, seq=5/1280, ttl=114 (request in 71)

```
▸ Frame 58: 50 bytes on wire (400 bits), 50 bytes captured (400 bits) on interfa
▸ Ethernet II, Src: PCSSystemtec_d1:5f:c0 (08:00:27:d1:5f:c0), Dst: Fortinet_09:
▸ Internet Protocol Version 4, Src: 192.168.73.155, Dst: 8.8.8.8
▾ Internet Control Message Protocol
    Type: 8 (Echo (ping) request)
    Code: 0
    Checksum: 0x2d35 [correct]
    [Checksum Status: Good]
    Identifier (BE): 51966 (0xcafe)
    Identifier (LE): 65226 (0xfeca)
    Sequence Number (BE): 2 (0x0002)
    Sequence Number (LE): 512 (0x0200)
    [Response frame: 59]
  ▾ Data (8 bytes)
      Data: 6532333031373630
      [Length: 8]

0000  00 09 0f 09 00 19 08 00  27 d1 5f c0 08 00 45 00   ····· '·_···E·
0010  00 24 fc 5d 40 00 40 01  24 28 c0 a8 49 9b 08 08   ·$·]@·@· $(··I···
0020  08 08 08 00 2d 35 ca fe  00 02 65 32 33 30 31 37   ····-5·· ··e23017
0030  36 30                                              60

Data (data.data), 8 bytes                    Packets: 875 · Displayed: 10 (1.1%)
```

Sound Settings


65 25.064911706 192.168.73.155 8.8.8.8 ICMP 50 Echo (ping) request id=0xcafe, seq=3/768, ttl=64 (reply in 66)
66 25.074951583 8.8.8.8 192.168.73.155 ICMP 60 Echo (ping) reply id=0xcafe, seq=3/768, ttl=114 (request in 65)
68 26.066613136 192.168.73.155 8.8.8.8 ICMP 50 Echo (ping) request id=0xcafe, seq=4/1024, ttl=64 (reply in 69)
69 26.077005101 8.8.8.8 192.168.73.155 ICMP 60 Echo (ping) reply id=0xcafe, seq=4/1024, ttl=114 (request in 68)
71 27.068812668 192.168.73.155 8.8.8.8 ICMP 50 Echo (ping) request id=0xcafe, seq=5/1280, ttl=64 (reply in 72)
72 27.080931897 8.8.8.8 192.168.73.155 ICMP 60 Echo (ping) reply id=0xcafe, seq=5/1280, ttl=114 (request in 71)
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: ICMP (1)
Header Checksum: 0x2373 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.73.155
Destination Address: 8.8.8.8
Internet Control Message Protocol
Type: 8 (Echo (ping) request)
Code: 0
Checksum: 0x2d34 [correct]
[Checksum Status: Good]
Identifier (BE): 51966 (0xcafe)
Identifier (LE): 65226 (0xfeca)
Sequence Number (BE): 3 (0x0003)
Sequence Number (LE): 768 (0x0300)
[Response frame: 66]
Data (8 bytes)
Data: 6532333031373630
0000 00 09 0f 09 00 19 08 00 27 d1 5f c0 08 00 45 00
0010 00 24 fd 12 40 00 40 01 23 73 c0 a8 49 9b 08 08
0020 08 08 08 00 2d 34 ca fe 00 03 65 32 33 30 31 37
0030 36 30
Data (data.data), 8 bytes
Packets: 907 · Displayed: 10 (1.1%)
Profile: Default